

Revision Notes for Class 8 Maths Chapter 7 Proportional Reasoning

Proportional reasoning helps us identify patterns when things change, especially when sizes increase or decrease but keep the same shape. For example, resizing digital images so that the width and height change by the same factor will keep images looking similar, while changing just one dimension will distort them. Measuring and comparing changes using ratios gives a clear way to check if objects or quantities remain in proportion.

Ratios and Proportion

The idea of a ratio is a central tool in proportional reasoning. A ratio like 60:40 means that for every 60 units of the first quantity, there are 40 units of the second. In everyday life, ratios help us compare things directly, such as sizes in images or mixtures for food and drinks. When two ratios can be written in the same simplest form, they are proportional, which means their values compare in the exact same way.

Simplifying Ratios

To find if two ratios are proportional, we can simplify both by dividing each part by their highest common factor (HCF). For example, simplifying 60:40 by dividing both terms by 20 gives 3:2. If another ratio also simplifies to 3:2, then both are proportional. Not all ratios simplify to the same numbers, which indicates they aren't proportionally related.

Checking for Proportionality in Real Situations

In daily life, we often check proportionality using ratios. For instance, making drinks with the same taste each time means keeping proportions of ingredients the same. If Kesang prepares 6 glasses of lemonade with 10 spoons of sugar, and she wants to make 18 glasses, she should multiply both numbers by 3 to keep the same ratio, so she'll use 30 spoons of sugar.

Similarly, if two people build walls of different lengths but use cement in the exact same proportion, both walls will have equal strength. However, ratios don't always stay

constant; adding or subtracting the same number to both quantities in a ratio will change the ratio.

Using Proportion to Find Missing Values

To solve real-life problems where one value is missing, we use the “Rule of Three.” If we know three numbers in two proportional ratios (like 120 students need 15 kg of rice, so 80 students will need how much?), we write the proportion as $120:15::80:x$. Solving gives $x = (15 \times 80)/120 = 10$ kg. The cross-multiplication method uses the formula $ad = bc$ to find unknowns in such cases.

Division in a Given Ratio

Sharing or dividing things in a specified ratio is common. To divide a given number into two parts in the ratio $m:n$, multiply the total number by $m/(m+n)$ and $n/(m+n)$ for the first and second part respectively. For example, dividing 42 counters in the ratio 4:3, first part = $42 \times 4/7 = 24$; second part = $42 \times 3/7 = 18$. This method helps in fair distribution of profits, mixtures, or resources.

A similar approach works for profit sharing. If Prashanti invests ₹75,000 and Bhuvan ₹25,000 in a business and total profit is ₹4,000, the profit is shared in 3:1, so Prashanti gets ₹3,000 and Bhuvan ₹1,000.

Mixing and Adjusting Ratios

Sometimes, we need to adjust an existing ratio to reach a new target. For example, if 40 kg of a sand-cement mixture is in the ratio 3:1 and we want to change it to 5:2, calculate how much to add or subtract for the new ratio. This skill is practical for recipes and construction.

Applications and Exercises

Students often practice proportional reasoning with tasks like matching similar shapes, completing tables of proportional values, or working out fair divisions. Exercises can include dividing an amount of money in a particular ratio, calculating the ingredients

needed for food recipes, or analyzing classroom data for student-teacher ratios. These practical exercises reinforce understanding.

Unit Conversions and Proportion

Many proportional problems require unit conversions. For example, to compare areas, students should know that 1 square metre is about 10.764 square feet, and 1 acre is 43,560 square feet. Volume units like 1 litre = 1,000 mL or 1,000 cc are often needed. Temperature conversions are also proportional: $^{\circ}\text{F} = (9/5) \times ^{\circ}\text{C} + 32$, and $^{\circ}\text{C} = (5/9) \times (^{\circ}\text{F} - 32)$.

Summary of Key Formulae and Points

- A ratio $a : b$ means for every 'a' units of one thing, there are 'b' units of another.
- Ratios are proportional if they become equal in simplest form or if $ad = bc$.
- To divide x into $m : n$, first part = $(m \times x)/(m + n)$, second part = $(n \times x)/(m + n)$.
- For cross-multiplication, if $a : b :: c : d$ then $ad = bc$.
- Exercises like market surveys or dividing mixtures help to apply these ideas in daily life.

Practice and Critical Thinking

Students are encouraged to solve problems that test whether two statements are in proportion, complete missing values in proportional ratios, and analyze real differences in cost or efficiency based on ratios (e.g., cost per kg in different brands or efficiency of different fertilizers). Applying proportional reasoning helps deepen understanding across mathematics and daily life.